

INFORMATION LEAKS & ENCRYPTION

GROUP 10 AND 11

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WHAT ARE INFORMATION LEAKS?

Information leakage happens when a system that is supposed to be private reveals information to people who are not allowed to see it. It usually occurs when the system's default settings share too much detail about how the website or server is built.



WHAT IS ENCRYPTION

Encryption is a process that takes normal data and "scrambles" it using a Secret Key. Once scrambled, the data is unreadable to anyone who does not have the Decryption Key. This keeps your information private and safe from hackers.

SYMMETRIC ENCRYPTION

Symmetric encryption is a cryptographic method that uses a single, shared secret key to both encrypt and decrypt data, making it highly efficient for processing large volumes of information. It is faster than asymmetric encryption but requires secure, out-of-band key sharing to ensure only authorized parties can access the data.

ASYMMETRIC ENCRYPTION

uses a pair of mathematically related keys—a public key for encryption and a private key for decryption—to secure data, allowing anyone to encrypt messages that only the holder of the private key can decrypt. It is widely used in HTTPS, secure email, and digital signatures

HASH FUNCTIONS

Hash functions are deterministic algorithms that map variable-length data into fixed-size, unique, or near-unique output values (digests). Used for rapid data retrieval, file integrity checks, and digital signatures, they must be fast to compute and collision-resistant. Common algorithms include SHA-256 and MD5, widely used in cryptography and database indexing.

MESSAGE AUTHENTICATION CODES

a short, symmetric cryptographic tag used to verify both the data integrity (has not been altered) and authenticity (originates from a trusted sender) of a message

CIPHER SUITE

A cipher suite is a set of cryptographic algorithms—including key exchange, authentication, bulk encryption, and message authentication code (MAC)—used to secure network connections via Transport Layer Security (TLS) or SSL. During the TLS handshake, the client and server negotiate to pick the strongest mutually supported suite to protect data integrity and privacy, such as AES-GCM or ChaCha20.

THANK YOU